UN-PC-002

Full Restoration, Data Evaluation, Linux Conversion, DigiDMS Access, and Remote Support Guide

Windsor Medical Center - Union Office

Source limitation

This guide is based only on the information, photographs, commands, corrections, testing, and decisions contained in this conversation. It does not use information from other conversations. No photographs are included.

Security redaction

Passwords, remote-access IDs, private home addresses, and other reusable credentials shown during the conversation are deliberately not reproduced. Any credential that appeared in a photograph should be treated as exposed and changed before final deployment.

Prepared from the complete project history in this conversation

Final corrected hardware record: HP Pavilion p7-1225 with PNY CS900 1 TB 2.5-inch SATA SSD

Contents

1. Scope, source rules, and corrections

This document reconstructs the entire UN-PC-002 project from the first physical installation questions through the final corrected configuration and the plan to return the computer to the Union office. It is intentionally chronological. It includes the successful steps, the failed attempts, the wrong assumptions that were later corrected, the retesting, and the final state of the machine.

The conversation included information from the medical-office equipment inventory and the office IT policy. Those documents were used only where they appeared in this conversation and where they supported the facts being recorded. No information from another chat or unrelated project is used here.

Three corrections control the entire guide

1) The workstation is UN-PC-002, not UN-PC-001. UN-PC-001 remained the fully functional primary Union workstation. 2) The computer is an HP Pavilion p7-1225, not an HP Compaq 6000 Pro. 3) The replacement drive is a PNY CS900 1 TB 2.5-inch SATA SSD, not a 500 GB SSD. Earlier drafts that used the wrong workstation ID, model, or SSD size are superseded by these corrected facts.

The guide distinguishes between facts directly demonstrated by photographs or command output and conclusions that were practical but not laboratory-proven. For example, the old Seagate drive was treated as failed because it was not usable in the old computer and would not identify correctly through the external dock. A final direct-SATA retest after removal was proposed but not performed because the user did not want to disturb the newly restored office computer.

2. Executive summary

UN-PC-002 began as an old Union office desktop that had reportedly been unusable for approximately three years. Its original 1 TB Seagate Barracuda 7200.12 mechanical hard drive was not functioning as a usable system drive. The computer was removed from the Union office and transported to the user's home for evaluation, data review, repair, and possible repurposing.

The computer was identified as an HP Pavilion p7-1225. A new PNY CS900 1 TB 2.5-inch SATA SSD was installed. The BIOS recognized the PNY drive, and a clean Lubuntu installation was completed. During setup, the CPU fan became loud and repeatedly changed speed. The case contained extensive dust. After the CPU fan, heatsink area, rear case fan, vents, and interior were cleaned with compressed gas and appropriate wiping of non-electronic surfaces, the noise stopped and the machine operated normally.

The old Seagate drive was removed from the drive cage and tested externally from a separate Windows laptop using a dual-bay SATA dock. The drive spun, but Windows did not identify it as a 1 TB Seagate. Disk Management and DiskPart instead reported an abnormal approximately 3.86 GB / 3950 MB USB disk with no volumes. CrystalDiskInfo could not read the Seagate or its SMART data. No patient, office, or business files could be accessed through the normal methods attempted in the conversation. The drive was then fully disassembled and retained as separated hardware components at home.

The rebuilt HP was configured with a new Windsor Medical Center administrator account named windsormed. Its final reported software base was Lubuntu on Ubuntu 24.04.4 LTS, kernel 7.0.0-28-generic, x86_64. Firefox initially failed because required Snap content packages were missing; the missing gnome-46-2404 and mesa-2404 content snaps were installed, after which Firefox opened successfully.

DigiDMS was not installed as a native Linux application. DigiDMS access was configured through a saved Remmina RDP profile that connects to the DigiDMS Windows environment. TeamViewer QuickSupport was used temporarily so DigiDMS support could assist with setup. RustDesk was then installed and tested so the user could remotely support the Union office computer from a personal laptop. The computer was prepared to return to the Union office as an active secondary Linux workstation.

3. Original system: identity, office role, and failure state

3.1 Correct workstation identity

Asset ID	UN-PC-002
Original location	Windsor Medical Center - Union office
Office relationship	This was not UN-PC-001. UN-PC-001 remained fully functional and continued as the primary Union Windows workstation.
Manufacturer	HP
Product family	HP Pavilion Series
Model	HP Pavilion p7-1225 PC
Product number	QW839AA#ABA
Serial number	MXU2050BF6
Original operating system	Genuine Windows 7 Home Premium
Original office status	Out of service / not functioning normally for approximately three years, according to the user's father

3.2 Original factory hardware

- Processor: AMD Quad-Core A8-3820 Accelerated Processor / APU, reported by BIOS at 2500 MHz.
- Memory: 8 GB DDR3. BIOS information showed 8192 MB DDR3 at 1333 MHz in dual-channel mode, using two 4096 MB modules.
- Graphics: AMD Radeon HD 6550D integrated graphics.
- Original storage: 1 TB mechanical hard drive.
- Optical drive: HP SuperMulti DVD burner; BIOS later showed an hpDVD-RAM GH80N device on SATA3.
- Wireless hardware: integrated Bluetooth and 802.11 b/g/n wireless LAN, according to the factory label.
- BIOS revision observed during the project: V7.15.
- Monitor used during the work: HP 2009m, 20-inch widescreen LCD, 1600 x 900, DVI/VGA.

3.3 Original failed storage device

Manufacturer	Seagate
Family	Barracuda 7200.12
Model	ST31000524AS
Capacity	1 TB
Spindle speed	7200 RPM
Serial number	6VPHM2VR
Part number	9YP154-304
Firmware	JC4B
Date code	12157
Date of manufacture	October 2011
Power label	+5 V 0.72 A; +12 V 0.52 A

The user's reason for examining the old Seagate was not merely hardware curiosity. Because it came from a medical-office computer, the user wanted to determine whether any patient or office information remained accessible before the drive was discarded or reused.

4. Objectives of the project

- Determine whether the old Union office computer could be made useful again.
- Determine whether the old Seagate drive contained any normally recoverable patient, office, or business files.
- Replace unreliable storage with a modern SSD.
- Install a lightweight operating system suited to the old hardware.
- Resolve any cooling, fan-noise, boot, browser, or package problems encountered during setup.
- Create an office account that did not use the user's personal name.
- Restore practical access to DigiDMS even though DigiDMS support staff were accustomed to Windows rather than Linux.
- Provide unattended remote IT support so routine problems could be handled without an unnecessary drive to Union.
- Document the old drive's chain of custody, failed recovery attempts, destruction, and the computer's return to office service.

5. Chronological restoration and conversion

5.1 Opening the computer and understanding the SATA connections

The first hardware question was how to connect the new SSD and whether to use the power lead labeled P3 or P4. The P3 and P4 plugs were two standard SATA power connectors from the power supply. Only one was required for the SSD. The practical choice was whichever connector reached the SSD cleanly without strain or an awkward bend.

- The wide connector on the SSD is SATA power and connects to one P3/P4 power plug.
- The narrow connector on the SSD is SATA data and connects to the motherboard with a SATA data cable.
- The connectors are keyed and should not be forced or installed backward.
- The 2.5-inch SSD was placed in the desktop's drive area/cage. Because a 2.5-inch SSD is smaller than a 3.5-inch hard-drive bay, it must be positioned securely enough that it cannot move or stress its cables during transport.

Final storage correction

The replacement drive was repeatedly described in early drafts as 500 GB. The user later corrected the record: the installed replacement is a PNY CS900 1 TB 2.5-inch SATA SSD.

5.2 BIOS and device detection

The HP Startup Menu and BIOS Setup Utility were used to verify that the new drive was visible to the motherboard. Under Device Configuration, the system reported:

```
Hard Disk
  SATA1   1.0 TB, PNY1TBSATASSD
CD-ROM
  SATA3   hpDVD-RAMGH80N
```

This confirmed that the PNY SSD was electrically connected and visible as a 1 TB SATA device. The DVD drive also remained present. This BIOS confirmation was important because it separated a storage-connection problem from an operating-system installation problem.

5.3 Booting the Lubuntu installer

The machine was booted into the Lubuntu installer. The installer displayed a warning that the computer did not satisfy every recommended installation requirement and that the system was not connected to the Internet. The installation was still allowed to continue.

Lubuntu was installed onto the new PNY SSD. The old Windows installation and user profiles were not migrated to the new drive. This created a clean separation between the old office storage and the rebuilt computer.

The final operating-system information later collected from the terminal was:

```
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.4 LTS
Release:        24.04
Codename:       noble
Kernel:         7.0.0-28-generic
Architecture:   x86_64
```

The desktop environment was Lubuntu/LXQt, while the base distribution identified itself as Ubuntu 24.04.4 LTS.

5.4 Fan noise during installation

While Lubuntu was running, the computer began making a loud whirring sound. The sound was traced to the CPU fan rather than the hard drive. The symptoms were carefully narrowed down:

- The CPU fan was visibly spinning rapidly.
- A slight vibration could be felt through the metal chassis.
- The fan smoothly slowed down and sped up repeatedly while continuing to spin.
- Lubuntu continued operating normally; the computer was not crashing or shutting down.
- The interior was visibly very dusty, especially around the CPU fan, heatsink, and rear case fan.

The open case and nearby air-conditioning vent did not eliminate the thermal problem. An open side panel does not remove dust packed into fan blades or heatsink fins, and it can disrupt the airflow path the chassis was designed to use. The nearby vent could make the surrounding air cool while the dust still acted as an insulating layer on the components.

5.5 Cleaning procedure and result

The user had a Max Pro compressed-gas electronics duster and cleaning wipes. The documented cleaning focused on the dust visible inside the computer. The important handling principles were: power the computer down, unplug it, avoid spraying liquid cleaner on electronic components, keep the compressed-gas can upright, use controlled bursts, and prevent fans from being driven at extreme speed by the air stream.

- Dust was removed from the CPU fan blades and the heatsink area behind the fan.
- Dust was removed from the rear case fan and surrounding grille.
- Loose dust was blown out of the chassis and vents rather than deeper into the machine.
- Wipes were used only where appropriate on non-powered, non-sensitive exterior or metal surfaces.

After the cleaning, the noise stopped. The fan returned to normal operation, and the computer continued functioning properly. The successful result strongly supported dust-restricted cooling as the immediate cause of the fan racing and speed cycling.

5.6 Removing the old Seagate from the drive cage

The old 3.5-inch Seagate was mounted in a removable metal drive cage. The removal was initially difficult because the cage was retained by a latch rather than by an obvious set of ordinary screws.

1. Shut the system down and disconnect power.

2. Disconnect the Seagate's SATA data and SATA power connections.
3. Locate and release the metal cage latch.
4. Lift or slide the entire drive cage out of the chassis.
5. Release the retaining rails/tabs that held the Seagate inside the carrier and remove the drive from the cage.

Once the latch was understood, the user successfully removed both the drive cage and the Seagate. The new PNY SSD remained as the system drive.

6. Old Seagate drive evaluation and data-recovery attempts

6.1 Reason for the evaluation

The Seagate came from an old medical-office workstation. The user wanted to know whether it still contained accessible patient or office information. The goal was inspection, not immediate formatting. This distinction became important because Windows later offered to initialize the disk.

6.2 External SATA dock and first detection issue

The Seagate was placed into a dual-bay SATA-to-USB docking station connected to a separate Windows laptop. At first, the dock's drive light did not illuminate and Windows did not show a second disk. The user then realized the hard drive had to be pushed firmly enough to seat on the dock's SATA connector. After reseating, the dock powered the drive and the Seagate spun.

Important separation of systems

The Windows laptop used for drive testing was a separate personal computer. The later PC App Store incident and Windows Defender scans occurred on that laptop, not on UN-PC-002.

6.3 Disk Management result

Windows Disk Management did not show a normal 1 TB Seagate. Instead, it showed an unknown Disk 1 with an abnormal capacity of approximately 3.86 GB. Windows prompted to initialize the disk as MBR or GPT.

There was an early discussion of choosing GPT because GPT is the normal modern partition style. That advice was corrected once the user clarified that the purpose was to inspect unknown existing contents. Initializing or formatting a disk can overwrite partition metadata, so the correct preservation step was to cancel the initialization prompt.

The user later attempted initialization anyway, but Windows returned:

```
Virtual Disk Manager
Access is denied.
```

No successful initialization or format was completed.

6.4 Device Manager result

Device Manager showed the external device as:

```
ST_M13FQ BL SCSI Disk Device
```

It did not identify itself as Seagate ST31000524AS or Barracuda 7200.12. The device properties also reported an abnormal capacity of 3950 MB and no volumes. The following identifiers were photographed during troubleshooting:

- Bus-reported description: ST_M13FQ BL SCSI Disk Device.
- Device instance path beginning with SCSI\DISK&VEN_ST_M13FQ&PROD_BL.
- Parent USB device beginning with USB\VID_0578&PID_0578.
- Location: Bus Number 0, Target Id 0, LUN 0.

These results showed that Windows could communicate with the USB/SCSI bridge path, but it was not receiving the real Seagate identity and 1 TB capacity.

6.5 DiskPart result

DiskPart was used from an elevated Command Prompt. The exact sequence was:

```
diskpart
list disk
select disk 1
detail disk
```

DiskPart reported Disk 0 as the laptop's approximately 476 GB internal disk and Disk 1 as an online 3950 MB USB disk. The detailed Disk 1 output showed:

```
ST_M13FQ BL SCSI Disk Device
Disk ID: 00000000
Type: USB
Status: Online
Path: 0
Target: 0
LUN ID: 0
Location Path: UNAVAILABLE
Current Read-only State: No
Read-only: No
Boot Disk: No
Pagefile Disk: No
Hibernation File Disk: No
Crashdump Disk: No
Clustered Disk: No
There are no volumes.
```

A healthy 1 TB drive would normally appear around 931 GB in Windows. The 3950 MB result was not consistent with the actual Seagate label.

6.6 CrystalDiskInfo result

CrystalDiskInfo was installed on the separate Windows laptop to read SMART data. It successfully read the laptop's own Samsung PM9C1a 512 GB NVMe SSD, proving that CrystalDiskInfo itself was functioning. The internal SSD showed Good (98%) health, approximately 529 power-on hours, 382 power cycles, approximately 9.1 TB read, approximately 8.46 TB written, and a temporary 67 C temperature while the laptop was under heavy scan and installation load.

The Seagate did not appear in CrystalDiskInfo at all, and no SMART data could be obtained from it.

6.7 Final practical conclusion and remaining uncertainty

The Seagate received power and spun, but it had not been usable in the original computer and it did not identify correctly through the external dock. The likely failure possibilities discussed were controller-board failure, firmware/service-area failure, internal head/preamp failure, or a dock/bridge communication problem.

A direct SATA retest after removal was proposed as the cleanest way to separate a drive failure from a dock failure. The user declined because the rebuilt office computer was already working and the user did not want to disturb its new setup. Therefore, the exact failed internal component was not proven. The final operational decision was to treat the Seagate as failed because it was not accessible through any normal method attempted and had already been nonfunctional in its original setting.

Data conclusion

The normal diagnostic tools used in the conversation could not expose any partition, file system, SMART record, patient record, or office file. This supports the statement that no information was recoverable through the ordinary methods attempted. It does not prove that every magnetic bit on an intact platter was forensically unrecoverable before the drive was dismantled.

7. CrystalDiskInfo download incident and Windows cleanup

7.1 How the wrong program was obtained

While obtaining CrystalDiskInfo, the user encountered a large advertisement on the legitimate Crystal Dew World download page that said PC App Store and Start Download. The advertisement visually resembled a download control. A separate file named Setup.exe was downloaded and run. The legitimate CrystalDiskInfo installer was a different file.

The Setup.exe properties showed approximately 4.46 MB, the description PC App Store Setup, and Fast Corporation LTD as the associated publisher. The legitimate CrystalDiskInfo 9.9.1 Aoi Edition installer was approximately 61.9 MB, identified Crystal Dew World in its properties, and had its own digital signatures.

7.2 Symptoms of the unwanted PC App Store

- A full-screen or always-on-top PC App Store overlay appeared.
- It displayed a misleading account/payment form with credit-card fields and a validation-charge message.
- It blocked or covered other applications and made Task Manager difficult or impossible to use normally.
- It resisted ordinary closing attempts.
- At shutdown, Windows reported that an application was preventing shutdown.
- The overlay returned after reboot.

The behavior was consistent with an intrusive potentially unwanted program or adware-style application. The conversation did not establish that it was a sophisticated Trojan.

7.3 Safe Mode removal and inspection

Because the overlay interfered with normal operation, the Windows laptop was started through Windows Recovery and into Safe Mode. In Safe Mode, the overlay did not control the desktop, allowing normal administrative work.

1. Open Windows Settings and review Installed apps.
2. Locate PC App Store, which appeared as a recently installed application from Fast Corporation LTD.
3. Uninstall the application.
4. Review Task Manager Startup apps for an obvious remaining startup item.
5. Inspect Downloads, Program Files, Program Files (x86), and the user's AppData folders for the installer and same-time residual files.

The user also inspected a newly created AppData\Local folder with a long hexadecimal-style name. It contained manifest.json and a newtab HTML file and was associated by timing with the unwanted installation. Normal folders such as D3DSCache and the PFU scanner-software folders were distinguished from the unwanted application.

7.4 Windows Security checks

Windows Security completed a quick scan on June 24, 2026. The screenshot recorded 29,525 files scanned, a 31-second scan, and 0 threats found. A full scan was then started; one screenshot showed 177,040 files scanned and approximately 21 minutes remaining. The final full-scan result was not shown in the conversation.

The user installed the legitimate CrystalDiskInfo while the full scan was running. This did not automatically invalidate the scan because real-time protection continued monitoring new files. The conversation nevertheless recommended allowing the full scan to finish and, if desired, running Microsoft Defender Offline afterward.

7.5 Final lesson from the incident

- Large Start Download advertisements are not necessarily the actual software download button.
- The legitimate CrystalDiskInfo installer and the PC App Store installer were separate files.
- Installer properties, file size, publisher, product name, and digital signature should be checked before execution.
- The PC App Store event happened on the separate Windows laptop used for diagnostics, not on UN-PC-002.

8. Physical destruction and teardown of the failed Seagate drive

8.1 Decision to dismantle the drive

After the drive failed to identify correctly and no normally accessible data could be obtained, the user decided not to spend more time on recovery or resale. Recycling through a public-works or electronics-recycling program was discussed, as was salvaging the strong magnets and other parts. The user chose to take the drive apart and keep the components.

8.2 Controller-board removal

The green printed circuit board on the underside was removed first. This board contained the SATA connector and the electronics that controlled power regulation, the spindle motor, cache, firmware, and communication. The board had a large insulating/thermal pad and components including a visible inductor marked 1R2. No obvious burn marks, exploded components, or heavy corrosion were visible in the photographs.

8.3 Opening the sealed drive enclosure

All visible Torx screws were removed from the top cover, but the cover still felt anchored. The remaining resistance was not simply the magnet. A concealed screw under a covered/label area was still holding the lid. After that hidden fastener was found and removed, the cover came off.

Once opened, the drive displayed its platter assembly, actuator arm, read/write head structure, voice-coil magnet assembly, spindle clamp, parking/ramp structures, and internal filter components.

8.4 Further disassembly

- The top actuator magnet assembly was removed. The user immediately observed how strong the neodymium magnet was.
- The spindle clamp and additional internal parts were removed.
- The screws were collected and stored in a plastic bag.
- The controller board, chassis, magnet pieces, actuator parts, platter assembly, filter pieces, and small hardware were retained at home as separated components.

The drive could no longer be reassembled and used as an ordinary hard drive after this teardown.

8.5 Black residue observed during teardown

The user noticed black or gray residue on the fingers after handling the open drive and platter area. The exact source was not established. The earlier conversation speculated about fine carbon or coating residue, but other plausible sources visible in the teardown were gasket material, adhesive, foam, black-painted parts, dust, or wear particles. It should therefore be recorded as unidentified internal residue rather than definitively described as platter lubricant or a silicon preservation package.

The user was advised to wash hands and avoid touching the eyes after handling the internal parts.

8.6 Documentation and data-destruction wording

The office inventory drafts documented the failed drive, unsuccessful normal recovery attempts, removal from service, complete mechanical disassembly, and off-site retention of separated parts. The most accurate wording is that the device was permanently rendered nonfunctional for ordinary operational use and that no data was accessible through the methods attempted.

Do not overstate the evidence

Because the conversation showed the platter being retained rather than professionally shredded or degaussed, the record should avoid claiming certified forensic sanitization unless additional destruction or certified disposal occurred later. The existing practical statement - no accessible data through normal methods and the drive mechanically dismantled - is supported.

9. Linux office account, system identity, and administrative setup

9.1 Creating the Windsor Medical Center account

The original Linux account used the personal name sanjev. Because the computer was returning to the office, a new office account was created from the terminal:

```
sudo adduser windsormed
```

The Full Name field was entered as Windsor Medical Center. A new /home/windsormed directory was created and the password was set. The account was then added to the sudo administrator group:

```
sudo usermod -aG sudo windsormed
```

The account was verified with:

```
id windsormed
```

The output showed UID 1001, GID 1001, and membership in the windsormed, sudo, and users groups. This confirmed that windsormed had administrative rights.

9.2 Removing the old personal account

The proposed cleanup command, to be run only while logged in as windsormed, was:

```
sudo deluser --remove-home sanjev
```

The conversation does not show a final screenshot confirming that this deletion was actually completed. It should therefore remain a deployment checklist item rather than be described as completed.

9.3 Computer hostname

The system hostname remained sanjev-p71225 in the terminal prompt. A professional inventory-aligned hostname such as un-pc-002 was recommended:

```
sudo hostnamectl set-hostname un-pc-002
sudo reboot
```

The conversation does not show final confirmation that the hostname was changed. The command hostname should be used after reboot to verify the final name.

9.4 System-information commands

```
lsb_release -a
uname -r
hostname
uname -m
hostnamectl
```

One typing mistake was documented: hostnamect1 was entered with the number 1 instead of hostnamectl with the letters ctl. The shell correctly reported that hostnamect1 was not found and suggested hostnamectl.

10. Firefox Snap failure and repair

10.1 Symptom

On the new windsormed account, clicking the Firefox icon did not open a browser window. Running firefox in the terminal produced the critical message:

```
Content snap GPU wrapper '/snap/firefox/8664/gpu-2404/bin/gpu-2404-provider-wrapper' not
found: ensure slot is connected
```

10.2 Diagnostic commands

```
snap list
snap connections firefox
snap interface content
```

Firefox 153.0-1 revision 8664 was installed, but its content plugs for gnome-46-2404 and gpu-2404 did not have matching provider slots. The system had older or unrelated content snaps but lacked the two required providers.

10.3 Failed repair attempt

The first repair attempt was:

```
sudo snap refresh
sudo snap connect firefox:gpu-2404
```

snap refresh reported that all snaps were current. The bare snap connect command failed because it attempted to use snapd as the provider and returned:

```
error: snap "snapd" has no "content" interface slots
```

This was a useful error because it showed that the missing issue was a provider package, not an out-of-date Firefox package.

10.4 Correct repair

```
sudo snap install gnome-46-2404
sudo snap install mesa-2404
```

After installation, the following command confirmed the correct providers:

```
snap connections firefox | grep -E 'gnome-46-2404|gpu-2404'
```

The connections then pointed to gnome-46-2404:gnome-46-2404 and mesa-2404:gpu-2404. Firefox opened successfully afterward.

10.5 Remaining nonfatal warnings

When Firefox first opened, the terminal printed additional warnings involving libpxbackend-1.0.so, a failed libgiolibproxy module load, permission parsing /etc/xdg/xdg-Lubuntu/gtk-3.0/settings.ini, and an IBus directory path under the Firefox Snap profile. Firefox nevertheless displayed its Welcome screen and was operational. The conversation did not document a later cleanup of those warnings.

11. DigiDMS access through TeamViewer QuickSupport and Remmina

11.1 Preparing information for DigiDMS support

The user contacted DigiDMS support and needed to explain that this was not a Windows workstation. The support-relevant system information was Ubuntu 24.04.4 LTS with the Lubuntu desktop, kernel 7.0.0-28-generic, x86_64 architecture, and the windsormed administrator account.

11.2 Launching TeamViewer QuickSupport on Linux

DigiDMS support provided a TeamViewer QuickSupport tar.gz package. It was first opened in the archive manager, but an executable cannot reliably be launched from inside the archive. The package was extracted to:

```
~/Downloads/teamviewerqs
```

The extracted folder contained config, doc, logfiles, tv_bin, and an executable launcher named teamviewer. The launcher already had execute permissions. It was started with:

```
cd ~/Downloads/teamviewerqs
./teamviewer
```

This allowed the support technician to assist temporarily.

11.3 Replacing the Windows mstsc expectation with Remmina

The support technician was familiar with Windows and attempted to use mstsc, the Windows Remote Desktop Connection command. Lubuntu does not provide mstsc. Remmina was used as the Linux RDP equivalent.

A saved Remmina connection profile named DigiDMS was created. It connected by RDP to the DigiDMS-hosted Windows environment on the server/port supplied by support. The Windows credentials included a domain and username. Those credentials and the password are intentionally omitted here.

11.4 Daily DigiDMS launch procedure

1. Log in to the windsormed Linux account.
2. Double-click the Remmina desktop icon.
3. If Lubuntu displays “This file Remmina seems to be a desktop entry,” choose Execute, not Open.
4. When Remmina opens, double-click the saved DigiDMS RDP profile.
5. Allow the Windows remote desktop to load and then use DigiDMS inside that remote Windows session.

Correct terminology

DigiDMS is not installed locally as a native Linux application. UN-PC-002 has a configured Remmina RDP connection that provides access to DigiDMS on a remote Windows environment.

11.5 Proper DigiDMS / RDP sign-out procedure

The DigiDMS technician specifically advised that users should not simply close the Remmina or remote-desktop window. Closing the window normally disconnects the RDP session but can leave the Windows user signed in on the server.

1. Finish and save all work inside DigiDMS.
2. Inside the remote Windows desktop, open Start and choose the user/profile menu.
3. Choose Sign out and wait for the Windows session to end.
4. Return to the Remmina client after Windows logs off.

5. If Remmina still shows an active connection, use Remmina Disconnect and then close the client.

11.6 Duplicate-session problem

Later, the RDP server presented a “Select a session to reconnect to” screen with two long-running sessions. Reconnecting to one and signing out seemed to return to the same chooser. Several possibilities were considered: stale disconnected sessions, the selected session being disconnected instead of logged off, or another office computer using the same shared RDP account.

The user then noticed a DigiDMS user associated with the East Windsor branch and concluded that another computer may have been actively using the shared DigiDMS/RDP credentials. This was a much safer explanation than assuming every listed session belonged to UN-PC-002.

Do not log off sessions blindly

A listed RDP session may belong to an employee working on another office computer and may contain unsaved work. Staff should confirm who is using DigiDMS, save work, and sign out normally. If stale sessions remain, DigiDMS Support should identify and clear them at the server level. Separate RDP accounts per workstation or user would reduce this ambiguity.

11.7 DigiDMS support responsibility

For DigiDMS application problems that are not caused by the local computer, operating system, network, printer/scanner configuration, Remmina, or another office-controlled setting, employees should contact DigiDMS Support first. Sanjev Rajaram should be brought in when DigiDMS Support cannot resolve the issue, identifies a local-computer cause, needs coordination, or the employee needs help carrying out support instructions.

12. RustDesk remote-support deployment

12.1 Why RustDesk was selected

The user wanted remote access without becoming dependent on TeamViewer licensing or additional commercial support subscriptions. RustDesk was chosen as the ongoing remote-support tool. Remmina remained the tool for connecting from Linux to the DigiDMS Windows server; RustDesk served the different purpose of allowing the user to control the Linux office computer from a personal laptop.

The conversation discussed eventual self-hosting. However, the setup completed here used the standard RustDesk ID/relay infrastructure. Therefore, the completed setup reduced TeamViewer dependence but did not yet eliminate all third-party infrastructure. Full independence would require a self-hosted RustDesk server, which was not completed in this conversation.

12.2 Download and installation

The official RustDesk GitHub release page showed version 1.4.9. Because the HP uses an AMD x86-64 processor, the correct package was the Ubuntu x86-64 .deb file, not ARM64.

The downloaded package was opened with QApt Package Installer. The terminal equivalent was:

```
cd ~/Downloads
sudo apt install ./rustdesk-1.4.9-x86_64.deb
```

12.3 First connection test

RustDesk was opened on both UN-PC-002 and the user’s personal Windows laptop. Each machine displayed its own RustDesk ID and one-time password. The office computer’s ID was entered on the personal laptop, and a remote-control session was successfully established. The IDs and passwords are not reproduced in this guide.

12.4 Security and permissions reviewed

The RustDesk Security page showed a broad custom permission set for administrator support. During the setup, the following permissions were enabled or shown enabled:

- Keyboard and mouse control
- Clipboard access
- File transfer
- Audio
- Camera
- Terminal
- TCP tunneling
- Remote restart
- Session recording
- Remote configuration modification

These are powerful permissions and should be restricted to the authorized IT administrator. Camera, terminal, TCP tunneling, file transfer, and remote configuration are not required for every help-desk task and should be reviewed periodically against the minimum access actually needed.

The password section was set to accept sessions using both a one-time password and a permanent password. A permanent password was to be set for unattended access. Two-factor authentication was visible but not enabled. “Keep screen awake during incoming sessions” was enabled. “Only allow connection if RustDesk window is open” was left off, which is necessary for unattended access.

12.5 Why Security could not initially be changed remotely

While an active RustDesk session was controlling the Linux PC, the Security tab could not initially be changed from within that same remote session. The settings were then opened directly on the Linux computer using the physical keyboard and mouse. This avoided locking out or changing the security of the active control session from inside itself.

12.6 Service, boot, and power requirements

The RustDesk General page showed the service running because the Service button offered Stop. For reliable unattended support, the service must start automatically when Linux boots. The final test should include a full reboot and a remote reconnection from the personal laptop without anyone clicking Accept.

- If UN-PC-002 is completely powered off, RustDesk cannot connect.
- If the computer is powered on at the login screen, remote access may work if the RustDesk service starts before login.
- The computer must have a working network connection. Ethernet is preferable for a stationary office workstation, although stable Wi-Fi can work.
- After a power outage, someone may need to press the power button unless the BIOS is configured to restore power automatically.
- Wake-on-LAN was discussed as a possible later enhancement but was not configured in this conversation.

12.7 Planned expansion

The user planned to add RustDesk to at least three additional office computers and to personal systems including a Linux Mint laptop, another Lubuntu computer, and the personal Windows laptop. The goal was remote-first troubleshooting for employees, with on-site visits reserved for power, hardware, cabling, network, or peripheral failures.

The conversation treated the free RustDesk setup as sufficient for the initial small deployment. Because licensing and service terms can change, that conclusion should be checked again before a larger or formal business rollout. RustDesk IDs may be stored in an internal IT list, but permanent passwords should be stored separately in a password manager.

12.8 Travel context and support model

Google Maps screenshots supplied by the user established the corrected travel comparison without repeating the private home address:

- Home to East Windsor office: 2.7 miles, approximately 6 minutes, using Southfield Road and Princeton Hightstown Road.
- Home to Union office: 44.3 miles, approximately 55 minutes under the shown conditions, via I-95 North and the Garden State Parkway North; the route includes tolls.

Employees work at Union on Tuesdays and Fridays and at East Windsor on the other weekdays. RustDesk is especially valuable for Union because a small software fix can otherwise require nearly two hours of round-trip driving. East Windsor remains easy to service physically, but remote access is still useful for fast diagnosis and determining whether an on-site visit is actually necessary.

13. Resale discussion and decision to retain/return the computer

After the restoration, resale value was discussed. The old platform's age limited its market value despite the new SSD, cleaning, and functioning Linux installation. A private-sale estimate in the conversation was roughly in the low hundreds of dollars, with a higher value only if bundled with a monitor, keyboard, and mouse. A $500 asking price was considered unrealistic compared with much newer refurbished computers.

The user stated that approximately $200 had been spent on the SSD and did not want to sell the SSD, computer, and monitor for an amount that merely recovered the drive cost. The machine was therefore more valuable as a working office or backup computer than as a resale item. The decision was made to return it to the father's Union office rather than sell it.

14. Final before-and-after comparison

Category	Before	After
Asset identity	Old Union office computer; initially confused in drafts with UN-PC-001.	Correctly documented as UN-PC-002. UN-PC-001 remained the primary working Union Windows PC.
Computer model	Initially misidentified in one draft as HP Compaq 6000 Pro.	HP Pavilion p7-1225, product QW839AA#ABA, serial MXU2050BF6.
Storage	Seagate Barracuda 7200.12 1 TB mechanical HDD, ST31000524AS, unusable.	PNY CS900 1 TB 2.5-inch SATA SSD as the new system drive.
Operating system	Original Windows 7 Home Premium installation on the old drive was inaccessible.	Clean Lubuntu desktop on Ubuntu 24.04.4 LTS, kernel 7.0.0-28-generic, x86_64.
User account	Personal Linux account named sanjev during setup.	Office administrator account windsormed, full name Windsor Medical Center.
Hostname	sanjev-p71225.	Recommended final hostname un-

		pc-002; completion must be confirmed.
Cooling	Heavy dust; CPU fan raced, slowed, and sped up; chassis vibration and whirring.	Interior cleaned; fan noise stopped; normal operation restored.
Browser	Firefox Snap would not launch because content providers were missing.	gnome-46-2404 and mesa-2404 installed; Firefox opened successfully.
DigiDMS	No working local office setup after conversion to Linux.	DigiDMS available through a saved Remmina RDP profile to the remote Windows environment.
Remote support	No established unattended support path.	RustDesk installed, connected, service running, and unattended-access settings prepared/tested.
Old drive data	Unknown because the old drive was inaccessible.	No files or SMART data accessible through attempted normal methods; drive dismantled and retained as separated parts.
Office status	Removed from Union and considered retired/non-production during restoration.	Prepared to return to Union as an active secondary Linux workstation.

15. Final corrected system record

Workstation ID	UN-PC-002
Location	Union office
Role	Secondary Linux office workstation; intended for the user's father and office administrative use
Manufacturer / model	HP Pavilion p7-1225 PC
Product / serial	QW839AA#ABA / MXU2050BF6
Processor	AMD Quad-Core A8-3820 APU, 2.5 GHz
Memory	8 GB DDR3, dual channel
Graphics	AMD Radeon HD 6550D integrated graphics
System drive	PNY CS900 1 TB 2.5-inch SATA SSD
Optical drive	HP SuperMulti DVD burner / hpDVD-RAM GH80N
Operating system	Lubuntu desktop based on Ubuntu 24.04.4 LTS, codename noble
Kernel	7.0.0-28-generic
Architecture	x86_64
Office account	windsormed; full name Windsor Medical Center; sudo administrator
Hostname observed	sanjev-p71225; recommended change to un-pc-002 was not confirmed
Browser	Mozilla Firefox Snap, repaired and functioning
DigiDMS access	Remmina RDP profile named DigiDMS; DigiDMS itself runs in a remote Windows environment
Temporary support tool	TeamViewer QuickSupport extracted and used during DigiDMS support setup
Ongoing remote support	RustDesk 1.4.9 installed and successfully connected from a personal Windows laptop

Cooling condition	Interior cleaned; prior fan noise resolved

16. Daily operating procedures for office staff

16.1 Starting the workstation

1. Power on the HP Pavilion and monitor.
2. Log in to the windsormed account.
3. Wait for the network connection and RustDesk service to become ready.
4. Do not modify administrator, network, RustDesk, or Remmina settings unless directed by the IT administrator.

16.2 Opening DigiDMS

1. Double-click the Remmina icon on the Lubuntu desktop.
2. When the desktop-entry dialog appears, click Execute.
3. Double-click the DigiDMS connection in Remmina.
4. Wait for the remote Windows desktop and DigiDMS login to appear.
5. Use DigiDMS normally inside the remote Windows session.

16.3 Ending DigiDMS correctly

1. Save and finish all DigiDMS work.
2. Use the Windows Start menu inside the remote session.
3. Choose the user/profile icon and select Sign out.
4. Wait until Windows completes the sign-out.
5. Back in Remmina, disconnect any remaining connection and then close Remmina.

Do not simply close the remote window, because that can leave a disconnected server session and cause the multiple-session chooser later.

16.4 Requesting IT help

1. Save open work and close or minimize unrelated patient information.
2. Report the affected workstation ID and the exact error message.
3. For application-side DigiDMS problems, call DigiDMS Support first.
4. Contact Sanjev Rajaram if the issue involves the local computer, Remmina, network, browser, printer/scanner settings, or if DigiDMS Support needs additional help.
5. Authorize the RustDesk session and avoid using the mouse/keyboard while remote troubleshooting is underway unless asked.
6. For power, hardware, cabling, internet, or physical-peripheral problems, an on-site visit may still be required.

17. Troubleshooting reference

17.1 Firefox does not open

```
firefox
snap connections firefox | grep -E 'gnome-46-2404|gpu-2404'
```

If the content connections are missing, verify that gnome-46-2404 and mesa-2404 remain installed. The original successful repair commands were:

```
sudo snap install gnome-46-2404
sudo snap install mesa-2404
```

17.2 DigiDMS shows multiple sessions

- Do not log off an unknown session until confirming that another employee is not using it.
- Ask staff at the other office to save and sign out normally.
- If stale sessions remain, ask DigiDMS Support to identify and clear them.
- A local reboot of Lubuntu does not necessarily remove sessions that live on the DigiDMS Windows server.

17.3 RustDesk cannot connect

- Confirm the office computer is powered on.
- Confirm the office has working Internet access.
- Confirm the RustDesk service is running and configured to start at boot.
- Confirm the permanent password has been set and has not changed.
- If the office network or power is down, an on-site visit is required.

17.4 Fan noise returns

- Observe whether the CPU fan or rear case fan is the source.
- Power down before opening the case.
- Check for new dust buildup, blocked heatsink fins, or a cable touching a fan blade.
- If the fan grinds, clicks, stalls, or wobbles after cleaning, replace the fan rather than continuing to run it.

17.5 Unknown external disk appears

If the purpose is to preserve or inspect possible data, do not initialize, clean, partition, or format the disk until its identity and capacity are confirmed. Use Disk Management, Device Manager, DiskPart, and a SMART tool first. A normal 1 TB disk should not be treated as valid if Windows reports only approximately 3.86 GB.

18. Mistakes, retesting, corrections, and lessons learned

- **Workstation identity:** The project computer was at times confused with UN-PC-001. The user corrected that it was UN-PC-002 and that UN-PC-001 remained fully functional.
- **Computer model:** An early inventory draft called it an HP Compaq 6000 Pro. The factory label proved it was an HP Pavilion p7-1225.
- **SSD capacity:** The replacement SSD was repeatedly called 500 GB. The user's final correction established a PNY CS900 1 TB 2.5-inch SATA SSD.
- **DigiDMS wording:** It is inaccurate to say that DigiDMS was installed natively on Linux. Access was configured through Remmina to a remote Windows environment.
- **Disk initialization:** GPT was initially discussed before the preservation goal was clear. The correct action for an unknown possible-data drive was to cancel initialization. A later initialization attempt failed with Access denied, so no successful format occurred.
- **Drive-versus-dock diagnosis:** At different points the abnormal 3.86 GB device was blamed on the dock or the Seagate. The final practical conclusion was a failed Seagate, but a post-removal direct-SATA control test was not completed, so the exact component failure remained unproven.
- **CrystalDiskInfo download:** A misleading PC App Store advertisement was clicked. The unwanted Setup.exe was separate from the legitimate CrystalDiskInfo installer.
- **Malware classification:** The behavior was intrusive and unsafe, but the conversation did not prove a sophisticated Trojan. Windows Security's quick scan found zero threats after removal.

- **Firefox repair:** A bare snap connect command failed because the provider snaps were missing. Installing gnome-46-2404 and mesa-2404 was the successful fix.
- **Hostname command typo:** hostnamect1 used the number 1 and failed. hostnamectl is the correct command.
- **TeamViewer package:** The TeamViewer QuickSupport archive was initially opened rather than extracted. It had to be extracted and launched with ./teamviewer.
- **Windows RDP command:** The support technician expected mstsc, which does not exist on Lubuntu. Remmina was the Linux RDP client.
- **RDP sign-out:** Closing the Remmina window can disconnect rather than log off. Windows Sign out must be used inside the remote session first.
- **Duplicate sessions:** The sessions were initially treated as stale sessions from the new computer. Later evidence suggested another office computer using the shared account could own one of them.
- **Drive teardown residue:** The black residue was speculatively described as platter coating. The source was not established and could have been gasket, adhesive, foam, painted material, or dust.
- **Return date:** The conversation used July 25, 2026 in one inventory update and later showed July 27 setup/testing with a planned return "tomorrow." The actual physical return date must be confirmed in the official record.

19. Inventory, retired-equipment, and IT-policy updates

19.1 Active workstation inventory

The Workstations and Computers section should contain a corrected UN-PC-002 entry using the final hardware and software record in Section 15. The Union workstation count should be increased to include both UN-PC-001 and UN-PC-002 once UN-PC-002 is physically returned and placed in service.

19.2 Retired drive record

The retired-equipment log should continue to record the Seagate drive separately from the computer. The hard drive remained retired and dismantled even though the computer chassis returned to active service. The record should include the Seagate model, serial, failed normal recovery attempts, removal, disassembly, off-site retention of parts, and the staff member responsible.

19.3 Return-to-service record

The old repurposed-workstation narrative remained historically true for the period when the computer was removed from the office and used as a non-production Linux machine. A later addendum was required to state that the same rebuilt computer was returning to Union as an official office asset with new storage, a clean Linux installation, Remmina/DigiDMS access, and RustDesk support.

19.4 Remote-support IT-policy addendum

The IT policy was updated conceptually to authorize RustDesk for remote technical support, prohibit employees from altering remote-access settings or sharing credentials, require staff to save work and close unrelated PHI before a support session, and preserve on-site service for problems that require physical access. The policy also clarified that DigiDMS Support is the first contact for DigiDMS application-side issues, with Sanjev providing additional or local-computer help when needed.

20. Commands used during the project

20.1 Account and system information

```
sudo adduser windsormed
sudo usermod -aG sudo windsormed
id windsormed
lsb_release -a
uname -r
hostname
uname -m
hostnamectl
```

20.2 Proposed identity cleanup

```
sudo hostnamectl set-hostname un-pc-002
sudo reboot
sudo deluser --remove-home sanjev
```

The hostname change and old-account deletion were recommended, but completion was not shown in the conversation.

20.3 Firefox Snap diagnosis and repair

```
firefox
snap list
snap connections firefox
snap interface content
sudo snap refresh
sudo snap install gnome-46-2404
sudo snap install mesa-2404
snap connections firefox | grep -E 'gnome-46-2404|gpu-2404'
```

20.4 TeamViewer QuickSupport

```
cd ~/Downloads/teamviewerqs
ls -l
./teamviewer
```

20.5 RustDesk installation

```
cd ~/Downloads
sudo apt install ./rustdesk-1.4.9-x86_64.deb
```

20.6 Windows drive inspection

```
diskpart
list disk
select disk 1
detail disk
```

21. Final deployment checklist and unresolved confirmations

- Confirm the actual physical return-to-service date and use that one date consistently in the inventory.
- Verify the computer's final hostname with hostname. If it still says sanjev-p71225, change it to un-pc-002 and reboot.
- Verify that the old sanjev Linux account has been removed only after confirming that windsormed has all required files and sudo access.
- Verify Firefox opens from the desktop after a cold boot, not only from a terminal session.

- Verify the nonfatal Firefox GTK/IBus warnings do not prevent normal use.
- Verify Remmina opens from its desktop icon by choosing Execute and that the DigiDMS profile connects.
- Verify staff know the proper Windows Sign out followed by Remmina Disconnect procedure.
- Confirm with DigiDMS Support whether shared RDP credentials are expected and whether separate accounts can be issued to prevent session conflicts.
- Confirm RustDesk has a strong permanent password and that it is stored in a secure password manager, not the general inventory.
- Reboot UN-PC-002 and confirm RustDesk reconnects from the personal laptop without local acceptance.
- Test RustDesk from a different network, not only while both computers are in the same house or LAN.
- Review RustDesk permissions and disable camera, terminal, TCP tunneling, file transfer, or remote configuration if they are not needed for normal support.
- Use Ethernet at Union if practical; otherwise verify stable Wi-Fi at the final desk location.
- Consider BIOS Restore on AC Power Loss if the office needs the computer to recover automatically after an outage.
- Do not store RustDesk passwords, DigiDMS passwords, or RDP passwords in the public inventory or employee-facing SOP.
- Rotate any password or reusable credential that appeared visibly in a photograph during this conversation.
- Update the Union workstation count in the inventory after deployment.
- Keep the dismantled Seagate parts secure and do not describe the destruction as certified forensic sanitization unless a stronger destruction step is completed and documented.

22. Conclusion

UN-PC-002 was not merely given a new operating system. It underwent a complete recovery and role change: the original failed storage was separated and evaluated, the computer received a new 1 TB SSD, the cooling system was cleaned, Lubuntu was installed and repaired, an office administrator account was created, Firefox packaging was fixed, DigiDMS access was rebuilt through Remmina, and remote IT support was established through RustDesk.

The project also produced several important operational lessons: preserve unknown drives before initializing them; verify installer identity before running it; distinguish local Linux software from remote Windows applications; log out of RDP sessions instead of merely closing windows; confirm whether shared sessions belong to another office computer; and separate public inventory facts from secret remote-access credentials.

At the end of the conversation, the computer was described as fully prepared for personal transport back to the Union office. Its final corrected identity is UN-PC-002, HP Pavilion p7-1225, with a PNY CS900 1 TB 2.5-inch SATA SSD and Lubuntu/Ubuntu 24.04.4 LTS. The original Seagate drive remained retired, dismantled, and stored as separated parts.

Appendix A. Compact final copy-and-paste workstation record

WORKSTATION ID / NAME:
UN-PC-002

Office Location:
Union

Manufacturer:
HP

Product Family:
HP Pavilion Series

Model:
HP Pavilion p7-1225 PC

Product Number:
QW839AA#ABA

Serial Number:
MXU2050BF6

Processor:
AMD Quad-Core A8-3820 APU @ 2.50 GHz

Memory:
8 GB DDR3 RAM

Graphics:
AMD Radeon HD 6550D integrated graphics

Storage:
PNY CS900 1 TB 2.5-inch SATA solid-state drive (SSD)

Operating System:
Lubuntu desktop / Ubuntu 24.04.4 LTS

Kernel:
7.0.0-28-generic

Architecture:
x86_64 (64-bit)

User / Role:
windsormed / Windsor Medical Center administrator account

DigiDMS Access:
DigiDMS is accessed through a saved Remmina Remote Desktop (RDP) profile connected to the DigiDMS Windows environment. DigiDMS is not installed as a native Linux application.

Remote IT Support:
RustDesk installed and configured for authorized remote troubleshooting by Sanjev Rajaram. On-site service remains available when a problem requires physical access.

Status:
Active / Returned to Union office service after complete storage replacement, operating-system reinstallation, internal cleaning, browser repair, DigiDMS remote-access setup, and remote-support configuration.

Notes:
The original Seagate Barracuda 7200.12 1 TB hard drive was removed after it failed to identify or provide accessible data through the normal diagnostic methods attempted. The old drive was mechanically dismantled and retained as separated parts. No original Windows installation, user profile, office application, or accessible patient/office data was transferred to the new SSD.

Appendix B. Compact DigiDMS / Remmina staff SOP

1. Double-click the Remmina desktop icon.
2. When asked what to do with the desktop entry, choose Execute.
3. Double-click the saved DigiDMS profile in Remmina.
4. Work inside the remote Windows/DigiDMS session.
5. When finished, use the Windows Start menu and choose Sign out.
6. After Windows signs out, disconnect the remaining Remmina connection and close Remmina.
7. Do not simply close the remote window, because that can leave a disconnected RDP session.

8. If multiple sessions appear, do not log them off blindly; first confirm whether another office computer or employee is using the shared account.
9. For DigiDMS application-side issues, call DigiDMS Support first. Contact Sanjev for local-computer, network, Remmina, printer/scanner, or unresolved support issues.

Appendix C. Compact RustDesk support statement

Authorized Windsor Medical Center computers may use RustDesk so Sanjev Rajaram can directly view and troubleshoot reported software or configuration problems. Staff should save open work and close or minimize unrelated patient information before a session. Remote support will generally be attempted first, especially for the Union office, while Sanjev remains available to come on site whenever a problem involves power, Internet service, hardware, cables, or another issue that cannot be corrected remotely.